

Pyxations: organizing, parsing, and analysing eye-movement data using Python

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Summary

Pyxations is a Python-based toolbox designed to unify the organization, parsing, and analysis of eye-movement data. It supports multiple common recording formats and detection algorithms, and offers integrated tools for preprocessing, visualization, and downstream analysis. By standardizing these steps and including a BIDS-inspired data structure, Pyxations facilitates reproducibility and makes it easier to compare results across tasks, devices, and studies.

Statement of Need

Looking at someone's eyes, searching for your keys or reading, are all active processes in which eye movements take a crucial role (Holmqvist et al., 2011; Land & Tatler, 2009; Rayner, 1998; Wade & Tatler, 2005). These movements include the saccades and fixations, focusing on the most relevant regions of the scene, but also smooth pursuit, microsaccades, or vergence. They are usually measured by optical eye-trackers, which consist of a camera collecting images from the eyes that ultimately provide the position and pupil size of both eyes on the scene. The cameras range from high-speed cameras (up to 2 kHz) to low-cost commercial webcams, and they also differ on the zoom applied and the inclusion of an IR source/filter. Finally, to go from the recording of the eyes' position to actual eye movements, it is necessary to detect such eye movements, for which there are many algorithms available, such as EyeLink (SR Research Ltd., 2021), Engbert and Mergenthaler (Engbert & Mergenthaler, 2006), REMoDNaV (Dar et al., 2021), among others.

Despite the central role of eye movements in vision research, there is currently no standardized pipeline or framework for organizing, parsing, and analyzing eye-tracking data equivalent to the Brain Imaging Data Structure (BIDS) (Gorgolewski et al., 2016). The heterogeneity of experimental paradigms, eye-tracking devices, and event-detection algorithms, combined with the variability introduced by calibration procedures and file formats, makes it difficult to move from raw recordings to reproducible and comparable results. This lack of standardization also complicates cross-study and cross-laboratory comparisons, as even similar experiments often yield data in incompatible formats or rely on different preprocessing pipelines.

In recent years, several open-source toolboxes have been developed to support eye-tracking data processing, including PyMovements (Krakowczyk & others, 2023), PyTrack (Ghose et al., 2020), and SPEED (Lozzi et al., 2025). Each of these offers valuable functionality within the growing ecosystem of open tools for gaze analysis, yet none fully address the combined challenges of dataset heterogeneity, reproducibility, and large-scale workflow automation that Pyxations is designed to solve. Below we summarize their key features and how Pyxations

extends or complements them.

PyMovements provides a modular Python interface for parsing, preprocessing, and analyzing eye-tracking data. It supports velocity-based event detection (fixations, saccades), data-quality metrics, and reproducibility guidelines. However, PyMovements is agnostic to data organization and format diversity: it does not enforce or automatically adapt to standardized folder hierarchies (e.g., BIDS-like structures), nor directly support data from multiple tracker vendors or legacy file types.

PyTrack is an end-to-end toolkit featuring fixation, saccade, and microsaccade extraction, estimation of multiple metrics (including pupil and blinks), and visualizations. Its graphical interface and built-in statistical tools make it accessible to non-programmers. Nonetheless, PyTrack is less suited to heterogeneous datasets, as it assumes a relatively uniform input structure. It also provides fewer options for integrating custom detection algorithms or preprocessing modules, making it harder to enforce reproducibility across diverse experimental pipelines.

SPEED (LabSoC Standardized Processing and Extraction of Eye-tracking Data) also focuses on lowering the entry barrier. However, SPEED is tailored to Pupil Labs' data and lacks flexibility.

Therefore, Pyxations was designed as a reproducible and extensible framework that unifies these complementary strengths while addressing their limitations. It is a Python-based toolbox designed to unify the organization, parsing, and analysis of eye-movement data. It supports multiple common recording formats and detection algorithms, and offers integrated tools for preprocessing, visualization, and downstream analysis. By standardizing these steps and including a BIDS-inspired data structure, Pyxations facilitates reproducibility and makes it easier to compare results across tasks, devices, and studies.

Research Impact Statement

Pyxations has already been deployed to standardize workflows in active research environments. It was utilized to process and analyze webcam-based eye-tracking data for a study on validation of an online antisaccade paradigm (Juantorena et al., 2023). Moreover, source code was also utilized for EyeLink data in a study on hybrid search strategies, which combined Bayesian and neural network models (Ruarte et al., 2025).

The library's versatility has been further validated through the successful harmonization of diverse datasets. It has been used to process data from GazePoint 3 trackers, in-house web-based eye-trackers, and driving simulation experiments (courtesy of collaborators mentioned in the acknowledgements), demonstrating its capacity to handle real-world heterogeneity beyond standard laboratory tasks. Furthermore, we are currently collaborating with three additional research groups to integrate Pyxations into their data analysis pipelines, establishing it as a growing standard for reproducible eye-tracking research.

Software Design

Pyxations is designed as a reproducible and extensible framework that unifies the complementary strengths of previous developments while addressing their limitations. Its core contributions can be summarized (Fig. 1) as follows:

Standardized dataset organization. Pyxations enforces a BIDS-inspired hierarchy (subjects, sessions, derivatives) (Gorgolewski et al., 2016), automatically harmonizing file naming and structure across vendors. This facilitates transparent sharing, version control, and collaborative reuse.

87 **Cross-format parsing and harmonization.** It supports multiple native formats (e.g., EyeLink
88 .edf/.asc, Tobii, WebGazer, and text-based legacy files) through a unified parsing API, reducing
89 friction when combining data from different acquisition systems.

90 **Calibration- and validation-aware preprocessing.** Pyxations directly parses calibration and
91 validation reports (e.g., EyeLink VALIDATION blocks), extracting per-session accuracy, offset,
92 and drift metrics. These can be used for automated exclusion or weighting of low-quality data.

93 **Flexible trial segmentation.** The framework accommodates multiple trialing paradigms: explicit
94 start and end timestamps, event-based markers, or fixed-duration trials. All of them with
95 overlap controls and regular-expression message matching. This flexibility enables consistent
96 parsing of diverse experimental logics without manual preprocessing.

97 **Declarative, provenance-aware workflow.** Every preprocessing operation (e.g., interpolation,
98 blink rejection, event detection) is logged automatically in machine-readable JSON recipes and
99 provenance sidecars. This ensures exact reproducibility of analysis pipelines across computing
100 environments.

101 **Scalability and performance.** Built on the Polars data engine (Vink, 2022), Pyxations executes
102 parallelized operations on large eye-tracking datasets, significantly outperforming traditional
103 pandas-based workflows (McKinney & others, 2020). This makes it suitable for multi-subject,
104 multi-session analyses typical in modern cognitive experiments.

105 **Visualization, statistics, and inspection tools.** In addition to standard gaze plots, Pyxations
106 includes dynamic scanpath visualizations, hierarchical data analysis (experiment, subject,
107 session, trial), per-trial calibration visualization, and task-specific visualization utilities tailored
108 to paradigms such as visual search. Multimatch metrics (Dewhurst et al., 2012) are also
109 embedded at the trial level to compare similarity between scanpaths.

110 **Integration and interoperability.** Pyxations can interface with existing libraries such as
111 PyMovements for event detection (Krakowczyk & others, 2023) or PyTrack for visualization
112 within its standardized processing pipeline (Ghose et al., 2020). It thus functions not as a
113 replacement but as an orchestration layer that harmonizes and scales the use of existing tools.

114 **Figure 1**

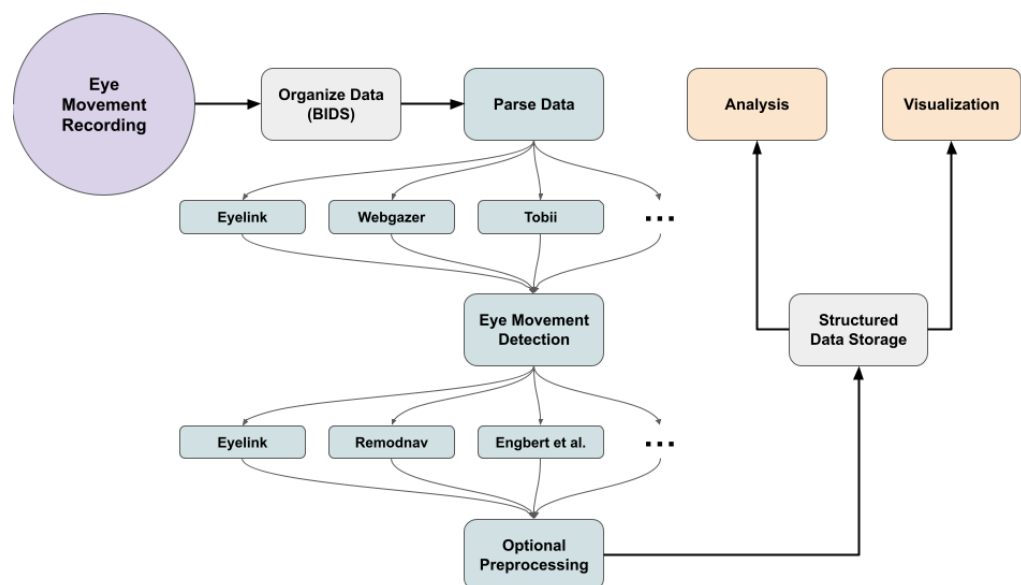


Figure 1: Preprocessing and analysis workflow showing parsing, event detection, preprocessing, and analysis stages.

115 In summary, Pyxations extends current open-source eye-tracking software by offering a scalable,
 116 provenance-aware, and calibration-informed framework that unifies parsing, preprocessing, and
 117 analysis within a standardized data structure. We want to keep building upon this framework
 118 which is why we decided to use design patterns for code scalability.

119 AI Usage Disclosure

120 We utilized generative AI tools (primarily ChatGPT-4o and Gemini 2.5 Flash) to a limited extent,
 121 specifically for generating individual methods based on established architectural decisions,
 122 creating unit tests, and drafting documentation strings. We did not employ autonomous
 123 AI agents or large-scale automated coding pipelines. Regarding the manuscript, these tools
 124 were used solely for linguistic refinement, such as typo spotting, grammar correction, and
 125 minor stylistic improvements. The core scientific content, figures, and bibliography were not
 126 generated by AI.

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